

LAB 5

HTML5 Semantic Tags, Form Validation & Multimedia

• Examples • Attributes • Output • Quick Revision

Part A — HTML5 Semantic Tags

Semantic elements give meaning to different parts of a webpage. They make the structure easier for students, developers, browsers and accessibility tools to understand.

1. <header>

Definition: Defines introductory or top content for a webpage or a section.

Purpose / Why we use it: Used for website titles, introductory content, logos or headings.

Syntax:

```
<header> ... </header>
```

Example:

```
<header>
  <h1>My Website</h1>
</header>
```

Expected Output: A heading appears as the introductory/top content.

Attributes:

Attribute	Belongs to	Meaning
id	<header>	Unique identifier.
class	<header>	Groups the element for CSS/JavaScript.
style	<header>	Adds inline CSS styling.
title	<header>	Shows extra information as a tooltip in supporting browsers.

Important Rules:

- It can contain headings, logos, introductory text and navigation-related content.

Common Mistakes:

- Do not confuse <header> with the whole HTML <head> element.

 **Remember:** header → introduction / top content

2. <nav>

Definition: Defines a section containing important navigation links.

Purpose / Why we use it: Used for menus such as Home, About, Services and Contact.

Syntax:

```
<nav> ... </nav>
```

Example:

```
<nav>
  <a href="#home">Home</a>
  <a href="#about">About</a>
</nav>
```

Expected Output: Navigation links appear.

Attributes:

Attribute	Belongs to	Meaning
id	<nav>	Unique identifier.
class	<nav>	CSS/JavaScript grouping.
aria-label	<nav>	Can describe the purpose of the navigation for accessibility.

Important Rules:

- Use it for major navigation, not every random link on the page.

Common Mistakes:

- Forgetting the href value on anchor links.

 **Remember:** nav → navigation

3. <main>

Definition: Defines the main unique content of the webpage.

Purpose / Why we use it: Keeps the primary page content separate from repeated site content such as navigation and footer.

Syntax:

```
<main> ... </main>
```

Example:

```
<main>
  <h2>Welcome</h2>
  <p>This is the main content.</p>
</main>
```

Expected Output: The main content appears.

Attributes:

Attribute	Belongs to	Meaning
id	<main>	Unique identifier.
class	<main>	CSS/JavaScript grouping.
style	<main>	Inline CSS styling.

Important Rules:

- A page normally has one main element for its primary content.

Common Mistakes:

- Putting repeated navigation or footer content inside main.

 **Remember:** main → main content

4. <article>

Definition: Defines self-contained content that can make sense on its own.

Purpose / Why we use it: Useful for blog posts, news articles, forum posts or product reviews.

Syntax:

```
<article> ... </article>
```

Example:

```
<article>
  <h2>HTML5 Basics</h2>
  <p>HTML5 provides semantic elements.</p>
</article>
```

Expected Output: An independent article appears.

Attributes:

Attribute	Belongs to	Meaning
id	<article>	Unique identifier.
class	<article>	CSS/JavaScript grouping.
datetime	Not an article attribute	Use <time> for publication/date information instead.

Important Rules:

- An article should represent a meaningful, independent piece of content.

Common Mistakes:

- Using article just because something is visually separated; meaning is more important than appearance.

💡 **Remember:** article → independent content

5. <section>

Definition: Groups related content under a common topic.

Purpose / Why we use it: Used to divide a page into meaningful topic-based sections, usually with a heading.

Syntax:

```
<section> ... </section>
```

Example:

```
<section>
  <h2>Our Services</h2>
  <p>Web Design and Development</p>
</section>
```

Expected Output: A related content section appears.

Attributes:

Attribute	Belongs to	Meaning
id	<section>	Unique identifier.
class	<section>	CSS/JavaScript grouping.
style	<section>	Inline CSS styling.

Important Rules:

- A section normally has a meaningful heading.

Common Mistakes:

- Using section only for visual spacing. Use semantic meaning, not appearance.

💡 **Remember:** section → related topic

6. <footer>

Definition: Defines footer information for a webpage or a section.

Purpose / Why we use it: Used for copyright, contact information, related links or author information.

Syntax:

```
<footer> ... </footer>
```

Example:

```
<footer>
  <p>© 2026 Yug Sardhara</p>
</footer>
```

Expected Output: Footer information appears.

Attributes:

Attribute	Belongs to	Meaning
id	<footer>	Unique identifier.
class	<footer>	CSS/JavaScript grouping.
style	<footer>	Inline CSS styling.
title	<footer>	Extra descriptive information.

Important Rules:

- A footer can belong to the whole page or to an individual section/article.

Common Mistakes:

- Thinking footer must always be the last HTML element; its meaning depends on the section it belongs to.

💡 **Remember:** footer → bottom/ending information

HTML Entity: ©

Definition:

© is an **HTML entity** used to display the copyright symbol © on a webpage.

Example:

```
<p>&copy; 2026 Yug Sardhara</p>
```

Expected Output:

© 2026 Yug Sardhara

Explanation:

The browser converts © into the copyright symbol ©.

Remember:

© → © (**Copyright Symbol**)

Common Use:

It is commonly used in the footer of a website to show copyright information.

```
<footer>
```

```
  <p>&copy; 2026 Yug Sardhara</p>
```

```
</footer>
```

Complete Semantic Page Example

```
<!DOCTYPE html>
<html>
<head>
  <title>Semantic Page</title>
</head>
<body>

  <header>
    <h1>My Website</h1>
  </header>

  <nav>
    <a href="#home">Home</a>
    <a href="#about">About</a>
  </nav>

  <main>
    <section>
      <h2>About Us</h2>
      <p>Welcome to our website.</p>
    </section>

    <article>
      <h2>Latest Article</h2>
      <p>HTML5 semantic tags give meaning to page structure.</p>
    </article>
  </main>

  <footer>
    <p>© 2026 Yug Sardhara</p>
  </footer>

</body>
</html>
```

Output: A webpage is organized into a header, navigation area, main content, section, article and footer.

 **Remember:** Think of semantic tags as labels that tell us what each part of the webpage means.

Part B — HTML5 Forms and Validation

HTML5 form controls collect information from users. Validation checks whether the entered information follows the rules.

7. <input type="email">

Definition: Creates an input intended for an email address.

Purpose / Why we use it: Used when the user must enter an email such as student@example.com.

Syntax:

```
<input type="email" id="email" name="email">
```

Example:

```
<label for="email">Email:</label>
<input type="email" id="email" name="email" required>
```

Expected Output: An email input box appears. The browser can perform basic email-format validation.

Attributes:

Attribute	Belongs to	Meaning
type	<input>	Specifies the input kind as email.
id	<input>	Unique identifier; connects the label using for.
name	<input>	Name used when form data is submitted.
required	<input>	Makes the field compulsory.
placeholder	<input>	Shows a temporary hint.

Important Rules:

- type=email provides basic browser validation; it is not a complete guarantee that an address actually exists.

Common Mistakes:

- Using type=text when email validation is expected.

 **Remember:** email → email address

8. <input type="tel">

Definition: Creates an input intended for a telephone/mobile number.

Purpose / Why we use it: Used for phone numbers. It does not automatically force a specific number of digits.

Syntax:

```
<input type="tel" id="phone" name="phone">
```

Example:

```
<label for="phone">Phone:</label>
<input type="tel" id="phone" name="phone" pattern="[0-9]{10}" required>
```

Expected Output: A phone input appears. The pattern makes exactly 10 digits required for validation.

Attributes:

Attribute	Belongs to	Meaning
type	<input>	Specifies telephone input.
id	<input>	Unique identifier.
name	<input>	Submitted field name.
pattern	<input>	Custom format rule.
required	<input>	Makes it compulsory.

Important Rules:

- Use pattern when you need a specific format such as exactly 10 digits.

Common Mistakes:

- Assuming tel automatically validates 10 digits. It does not.

💡 **Remember:** tel → telephone

9. <input type="date">

Definition: Creates a date input. Many browsers provide a date picker.

Purpose / Why we use it: Used for date of birth, joining date, appointment date, etc.

Syntax:

```
<input type="date" id="dob">
```

Example:

```
<label for="dob">Date:</label>
<input type="date" id="dob" min="2000-01-01" max="2026-12-31">
```

Expected Output: A date field/date picker appears.

Attributes:

Attribute	Belongs to	Meaning
type	<input>	Specifies date input.
id	<input>	Unique identifier.
min	<input>	Earliest allowed date.
max	<input>	Latest allowed date.
required	<input>	Makes it compulsory.

Important Rules:

- Date values use the HTML date format YYYY-MM-DD.

Common Mistakes:

- Typing a display format such as DD-MM-YYYY into the value attribute.

💡 **Remember:** date → date picker

10. <textarea>

Definition: Creates a multi-line text input area.

Purpose / Why we use it: Used for messages, addresses, comments and feedback.

Syntax:

```
<textarea> ... </textarea>
```

Example:

```
<textarea id="feedback" rows="5" cols="30" minlength="10" maxlength="200" placeholder="Write feedback..."></textarea>
```

Expected Output: A multi-line text box appears.

Attributes:

Attribute	Belongs to	Meaning
rows	<textarea>	Approximate visible number of text lines.
cols	<textarea>	Approximate visible width.
minlength	<textarea>	Minimum number of characters.
maxlength	<textarea>	Maximum number of characters.
placeholder	<textarea>	Temporary hint.
required	<textarea>	Requires a value.

Important Rules:

- textarea is not an input element; it has an opening and closing tag.

Common Mistakes:

- Writing the text between attributes instead of between <textarea> and </textarea>.

 **Remember:** textarea → multi-line text

11. required Attribute

Definition: Makes a form control compulsory before successful form submission.

Purpose / Why we use it: Used when the user must fill a field.

Syntax:

```
<input type="text" required>
```

Example:

```
<form>
  <input type="text" name="name" required>
  <input type="email" name="email" required>
  <button type="submit">Submit</button>
</form>
```

Expected Output: If a required field is empty, the browser normally shows a validation message and prevents successful submission.

Attributes:

Attribute	Belongs to	Meaning
required	<input>, <textarea>, <select>	Makes the control compulsory.

Important Rules:

- required is a Boolean attribute; writing required is enough.

Common Mistakes:

- Writing `required="false"` and expecting it to disable validation. Boolean attributes are presence-based.

💡 **Remember:** `required` → compulsory

12. pattern Attribute

Definition: Defines a custom regular-expression rule that the entered value must match during form validation.

Purpose / Why we use it: Useful for custom formats such as a 10-digit phone number or password rules.

Syntax:

```
<input type="text" pattern="...">
```

Example:

```
<input type="password" pattern="(?!.*[A-Z])(?!.*[0-9]).{8,}" title="8+ chars, uppercase and number" required>
```

Expected Output: When submitted, the browser checks whether the password matches the pattern.

Attributes:

Attribute	Belongs to	Meaning
pattern	<input>	Regular-expression rule.
title	<input>	Can explain the expected format to the user.
required	<input>	Makes the value compulsory.

Important Rules:

- pattern is generally used with text-like input types. It does not replace server-side validation.

Common Mistakes:

- Making a very complicated pattern without testing it.

💡 **Remember:** `pattern` → custom format/rule

13. type="number" + min + max

Definition: Creates a numeric input and can restrict its allowed value range.

Purpose / Why we use it: Useful for age, marks, quantity, rating and similar numeric values.

Syntax:

```
<input type="number" min="1" max="5">
```

Example:

```
<label for="rating">Rating:</label>
<input type="number" id="rating" min="1" max="5" required>
```

Expected Output: A numeric input appears. The allowed range is 1 to 5.

Attributes:

Attribute	Belongs to	Meaning
type	<input>	Specifies numeric input.
min	<input>	Smallest allowed value.
max	<input>	Largest allowed value.

step	<input>	Allowed increment.
required	<input>	Makes it compulsory.

Important Rules:

- min/max define the allowed range; they do not mean the input can only contain one specific value.

Common Mistakes:

- Confusing min/max with minlength/maxlength. min/max control values; minlength/maxlength control characters.

💡 **Remember:** number → numeric value | min/max → range

14. minlength and maxlength

Definition: Control the minimum and maximum number of characters allowed.

Purpose / Why we use it: Useful for names, feedback, comments and other text fields.

Syntax:

```
<input type="text" minlength="3" maxlength="30">
```

Example:

```
<textarea minlength="10" maxlength="200" required placeholder="Write feedback..."></textarea>
```

Expected Output: The browser validates the number of characters against the specified limits.

Attributes:

Attribute	Belongs to	Meaning
minlength	<input>, <textarea>	Minimum characters.
maxlength	<input>, <textarea>	Maximum characters.

Important Rules:

- These attributes count characters, not numeric value.

Common Mistakes:

- Using minlength when you actually want a minimum numeric value; use min for numeric ranges.

💡 **Remember:** minlength/maxlength → character count

Part C — Multimedia Tags

15. <audio>

Definition: Embeds sound/audio content in a webpage.

Purpose / Why we use it: Used to play music, recordings, podcasts or other audio.

Syntax:

```
<audio controls> ... </audio>
```

Example:

```
<audio controls loop>
  <source src="song.mp3" type="audio/mpeg">
  Your browser does not support audio.
</audio>
```

Expected Output: An audio player appears with controls and repeats after finishing.

Attributes:

Attribute	Belongs to	Meaning
controls	<audio>	Shows play/pause/volume controls.
autoplay	<audio>	Requests automatic playback; browsers may restrict audible autoplay.
loop	<audio>	Repeats the audio.
muted	<audio>	Starts without sound.
preload	<audio>	Hints how much media should be loaded before playback.
src	<audio>	Audio file URL; source can also provide it.

Important Rules:

- Use controls when students/users need to operate the audio.
- Autoplay with sound may be blocked by browsers.

Common Mistakes:

- Expecting autoplay to always work.
- Forgetting to provide a usable audio file path.

💡 **Remember:** audio → sound

16. <source>

Definition: Specifies a media file and its format for audio or video.

Purpose / Why we use it: Allows multiple media formats so the browser can select a supported one.

Syntax:

```
<source src="file.mp4" type="video/mp4">
```

Example:

```
<audio controls>
  <source src="song.mp3" type="audio/mpeg">
  <source src="song.ogg" type="audio/ogg">
</audio>
```

Expected Output: The browser chooses a supported audio source.

Attributes:

Attribute	Belongs to	Meaning
src	<source>	Path or URL of the media file.
type	<source>	MIME type of the media.

Important Rules:

- source is normally placed inside audio or video.

Common Mistakes:

- Putting <source> outside audio/video when it is intended to be a media source.

💡 **Remember:** source → media file

17. <video>

Definition: Embeds video content in a webpage.

Purpose / Why we use it: Used for lectures, demonstrations, tutorials, movies or other video content.

Syntax:

```
<video controls> ... </video>
```

Example:

```
<video controls poster="preview.jpg" width="640">
  <source src="movie.mp4" type="video/mp4">
  <source src="movie.webm" type="video/webm">
  Your browser does not support video.
</video>
```

Expected Output: A video player appears. The poster image is shown before playback.

Attributes:

Attribute	Belongs to	Meaning
controls	<video>	Shows playback controls.
poster	<video>	Image displayed before the video starts.
autoplay	<video>	Requests automatic playback; browsers may restrict it.
loop	<video>	Repeats the video.
muted	<video>	Starts without sound.
width	<video>	Displayed width.
height	<video>	Displayed height.
preload	<video>	Hints how much video data to load.

Important Rules:

- Use multiple source elements when you want to provide alternative formats.
- poster is an image, not the video itself.

Common Mistakes:

- Using poster as the video source.
- Assuming autoplay will always work.

💡 **Remember:** video → moving picture | poster → preview image

Complete Multimedia Example

```
<h2>My Audio</h2>
<audio controls loop>
  <source src="music.mp3" type="audio/mpeg">
</audio>

<h2>My Video</h2>
<video controls poster="poster.jpg" width="640">
  <source src="video.mp4" type="video/mp4">
```

```
<source src="video.webm" type="video/webm">
</video>
```

★ Quick Revision Summary for Students

Use this page for a fast revision before practical, viva or examination.

Topic	Remember in one line
<header>	Introduction / top content
<nav>	Navigation links
<main>	Main unique page content
<article>	Independent/self-contained content
<section>	Related content under a topic
<footer>	Footer/end information
type="email"	Email input
type="tel"	Telephone input; not automatically 10 digits
type="date"	Date input / date picker
<textarea>	Multi-line text
required	Compulsory field
pattern	Custom format rule
type="number"	Numeric input
min / max	Minimum and maximum VALUE
minlength / maxlength	Minimum and maximum CHARACTERS
<audio>	Audio player
controls	Shows media controls
autoplay	Requests automatic playback
loop	Repeats media
<source>	Media file + format
<video>	Video player
poster	Image shown before video starts

Important Differences

Do not confuse	Correct difference
min vs minlength	min = numeric value; minlength = number of characters
max vs maxlength	max = numeric value; maxlength = number of characters
tel vs number	tel = phone input; number = numeric value input
article vs section	article = self-contained content; section = related content group
poster vs source	poster = preview image; source = actual media file
required vs pattern	required = value must exist; pattern = value must follow a format

30-Second Memory Map

SEMANTIC → header | nav | main | article | section | footer

FORM → email | tel | date | textarea

VALIDATION → required | pattern | min | max | minlength | maxlength

AUDIO → audio | controls | autoplay | loop | source

VIDEO → video | controls | poster | source

 **Remember:** Before practical: first remember what each tag does, then remember the important attributes.

Student Self-Check

- ☐ Can I explain the difference between <article> and <section>?
- ☐ Can I explain why required is used?
- ☐ Can I explain pattern with one example?
- ☐ Can I distinguish min/max from minlength/maxlength?
- ☐ Can I explain tel vs number?
- ☐ Can I explain controls, autoplay and loop?
- ☐ Can I explain poster vs source?